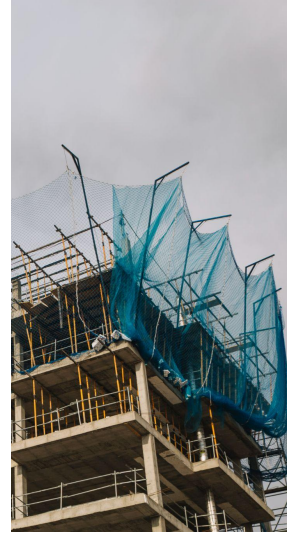


Implementing Software Architecture

Software Engineering Course – 5
2025-2026



Software Architecture

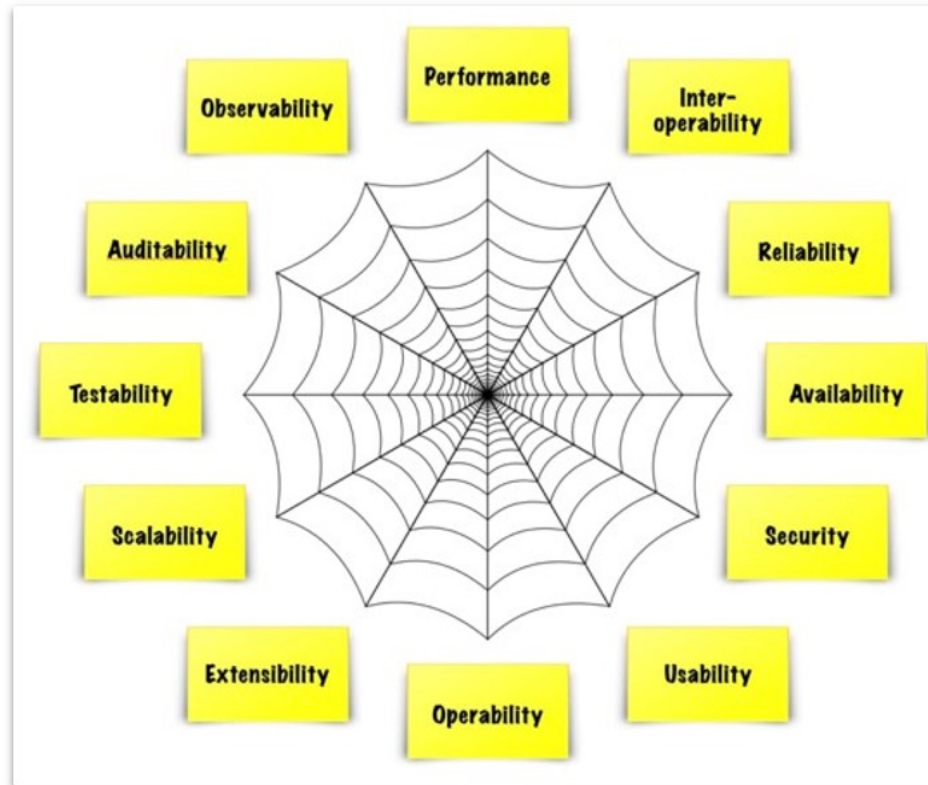
Building the right product

Functional Requirements

Building the product right

Non-Functional Requirements

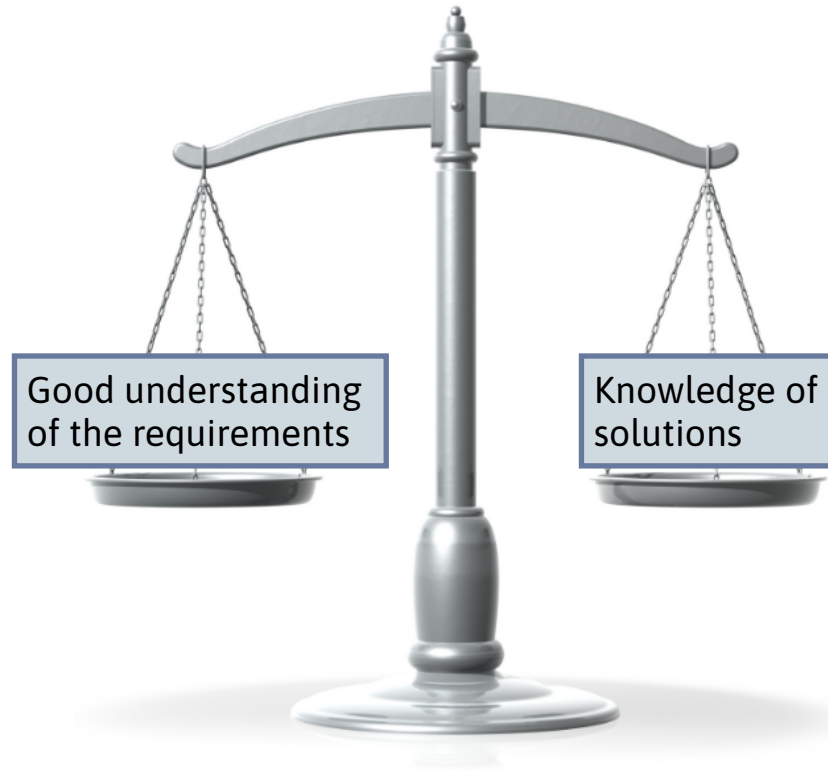
Non-Functional Requirements



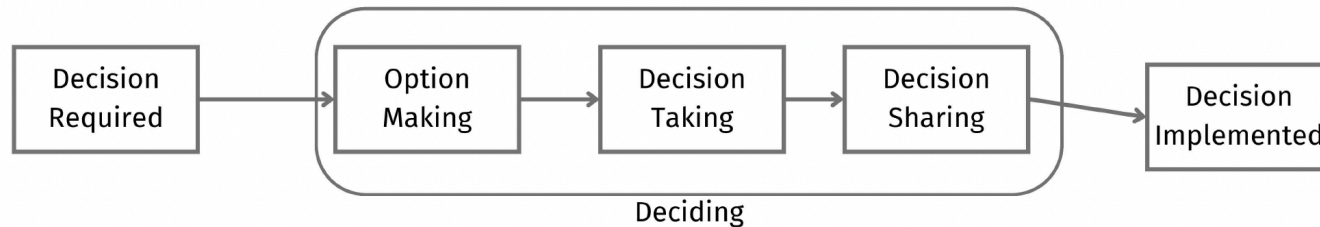
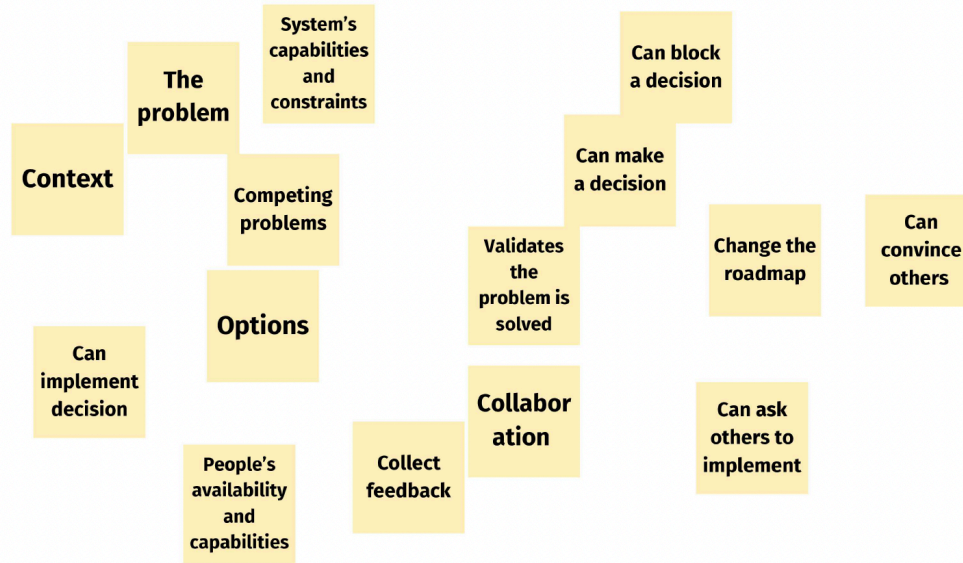
Trade-off Analysis



Trade-off Analysis



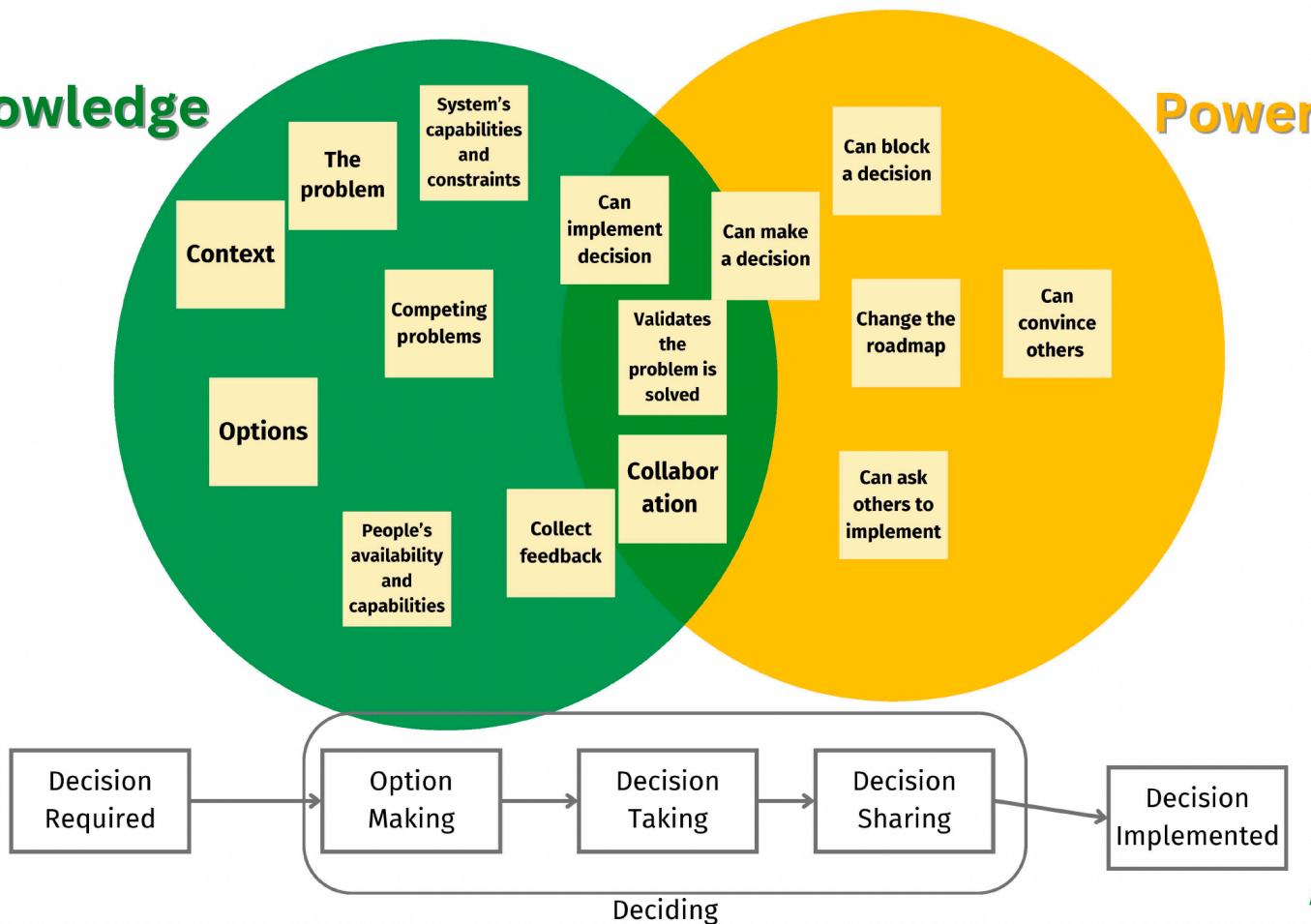
“Ingredients” of a decision

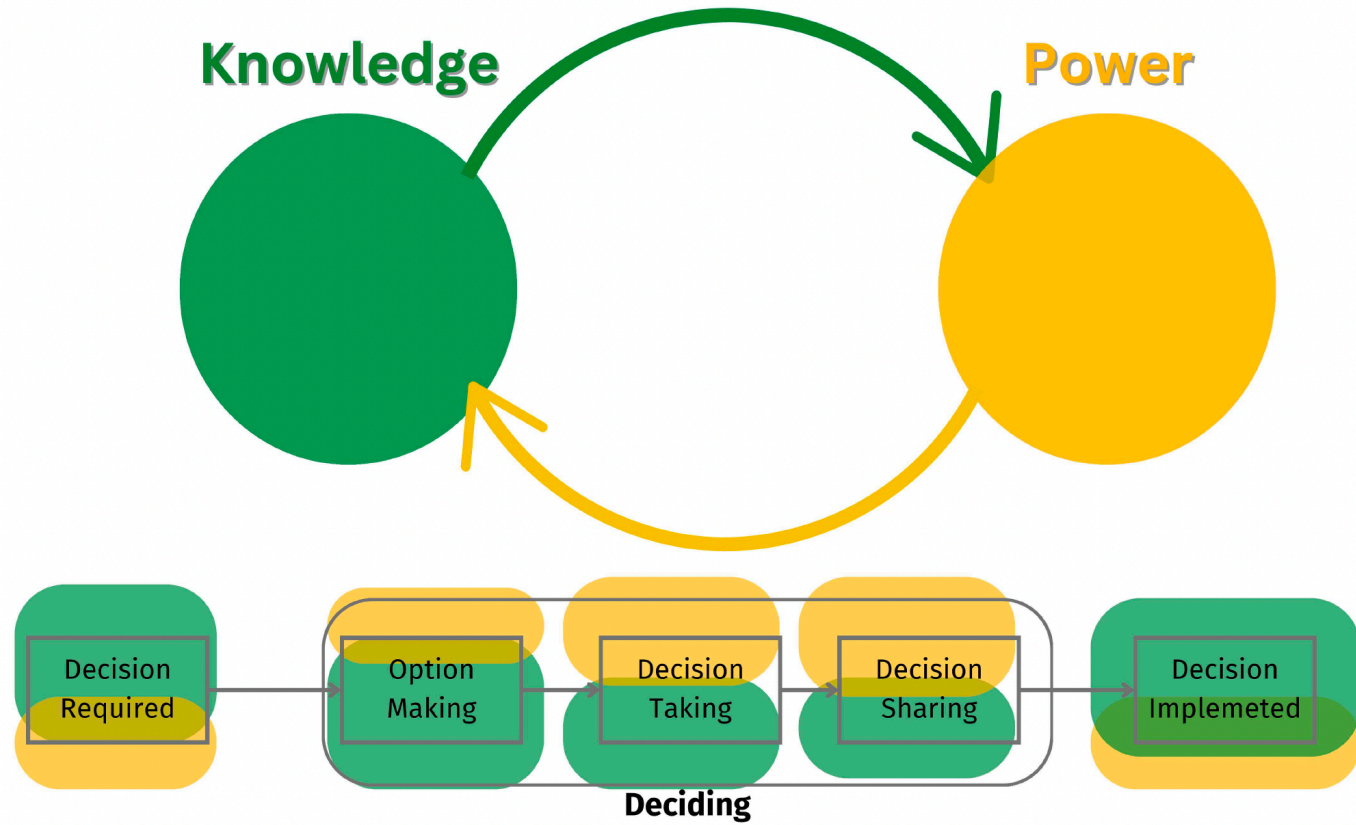




Knowledge

Power





Architecture Decision Records

☐	ADR Format
☐	TITLE
☐	Short description stating the architecture decision
☐	STATUS
☐	Proposed, Accepted, Superseded
☐	CONTEXT
☐	What is forcing me to make this decision?
☐	DECISION
☐	The decision and corresponding justification
☐	CONSEQUENCES
☐	What is the impact of this decision?
☐	COMPLIANCE
☐	How will I ensure compliance with this decision?
☐	NOTES
☐	Metadata for this decision (author, etc.)

Figure 19-1. Basic ADR structure

You can find in Teams the Chapter that tells you how to build an Architecture Decision Record

Architecture Decision Records

ADR 76. Asynchronous Pub/Sub Messaging Between Bidding Services

STATUS

Accepted

CONTEXT

The Bid Capture Service, upon receiving a bid from an online bidder or from a live bidder via the auctioneer, must forward that bid onto the Bid Streamer Service and the Bidder Tracker Service. This could be done using asynchronous point-to-point (p2p) messaging, asynchronous publish-and-subscribe (pub/sub) messaging, or REST via the Online Auction API Layer.

DECISION

We will use asynchronous pub/sub messaging between the Bid Capture Service, Bid Streamer Service, and the Bidder Tracker Service.

The Bid Capture Service does not need any information back from the Bid Streamer Service or Bidder Tracker Service.

The Bid Streamer Service must receive bids in the exact order they were accepted by the Bid Capture Service. Using messaging and queues automatically guarantees the bid order for the stream.

CONSEQUENCES

We will require clustering and high availability of the message queues.

Internal bid events will be bypassing security checks done in the API layer.

UPDATE: Upon review at the April 14th, 2020 ARB meeting, the ARB decided that this was an acceptable trade-off and no additional security checks would be needed for bid events between these services.

COMPLIANCE

We will use periodic manual code and design reviews to ensure that asynchronous pub/sub messaging is being used between the Bid Capture Service, Bid Streamer Service, and the Bidder Tracker Service.

NOTES

Author: Subashini Nadella

Approved By: ARB Meeting Members, 14 APRIL 2020

Last Updated: 15 APRIL 2020 by Subashini Nadella

Figure 19-5. ADR 76. Asynchronous Pub/Sub Messaging Between Bidding Services

Pitfalls of Software Architecture

- ✓ Don't let one person make or influence all the decisions
- ✓ Don't let reuse goals dictate bad decisions
- ✓ Don't let business decisions dictate your architecture. Build NFRs.
- ✓ Don't compromise quality for faster delivery
 - ✓ Create Technical Debt consciously
- ✓ Don't copy successful architecture. Design it yourself.
- ✓ Don't generalise if you aren't sure if it needs to be generalised.
- ✓ Build architecture incrementally
- ✓ Publish often.

What is Technical Debt? (It's not just "bad code")

Reckless

Prudence

Deliberate

Adapt culture
Agile Training

*"We don't
have time for
design!"*

*"We have to
ship now, and
schedule tech
debt solving"*

Follow-up
Set a Sprint for
solving Tech Debt

Inadvertent

Education
"Fail soon"

*"What's a
DNS?"*

*"Now we
know how we
should have
done it"*

Retrospective
How to do it even
better

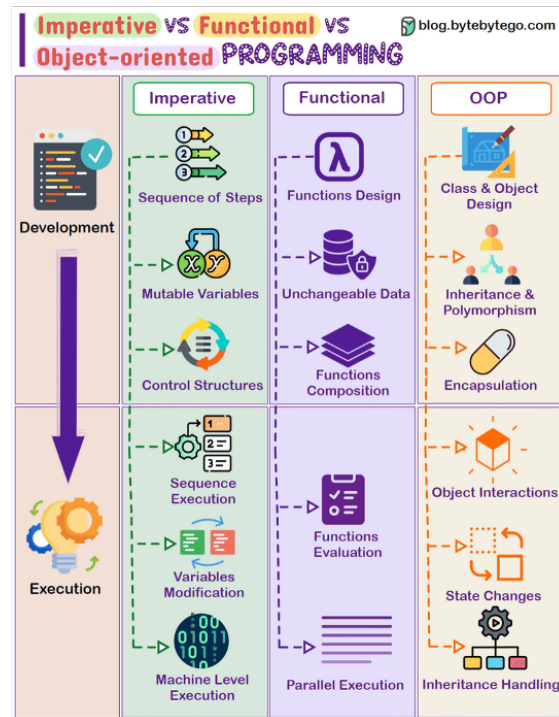
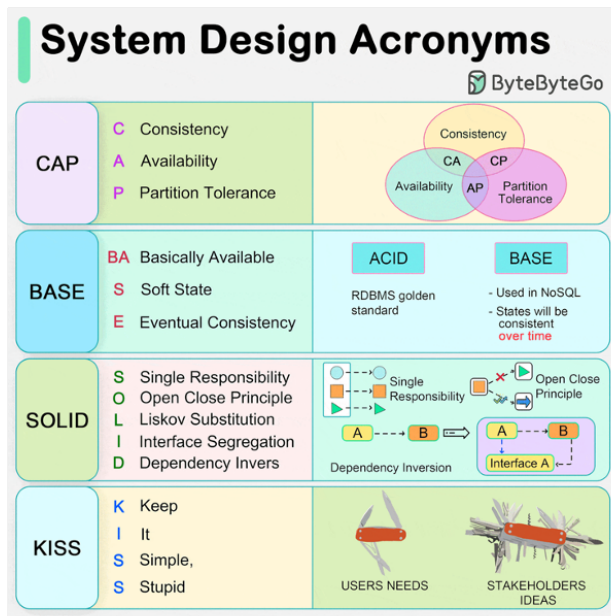
Managing Debt - The Architect's Repayment Plan

- **Make it Visible:** Add it to the backlog (e.g., "Refactor the payment module").
- **Justify It:** Use Architecture Decision Records (ADRs) to document why you are taking on debt.
- **Prioritize Repayment:** The "Boy Scout Rule" (leave the code cleaner than you found it), or dedicate a percentage of each sprint (e.g., 20% of time) to paying down debt.
- **Link it to Business Value:** "We need to refactor the database query system because it's currently costing us \$500/month in extra server fees and slowing down the user dashboard."

Learning resources for future Software Architects



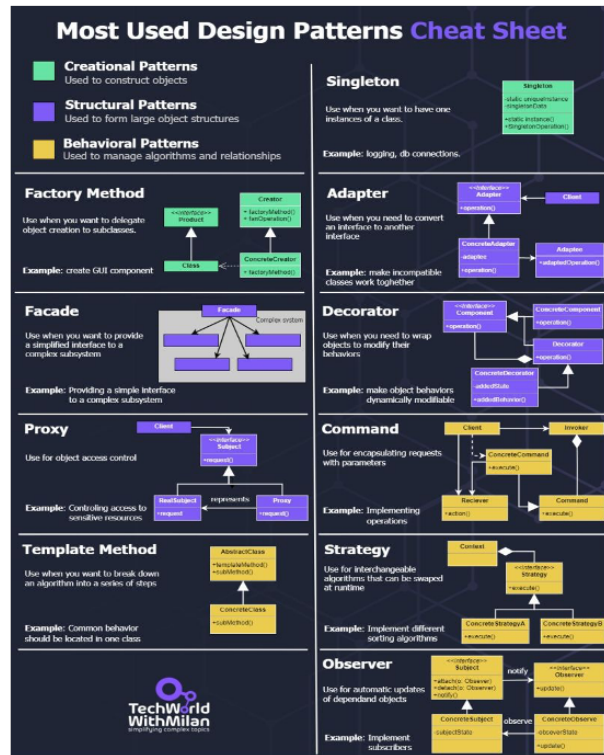
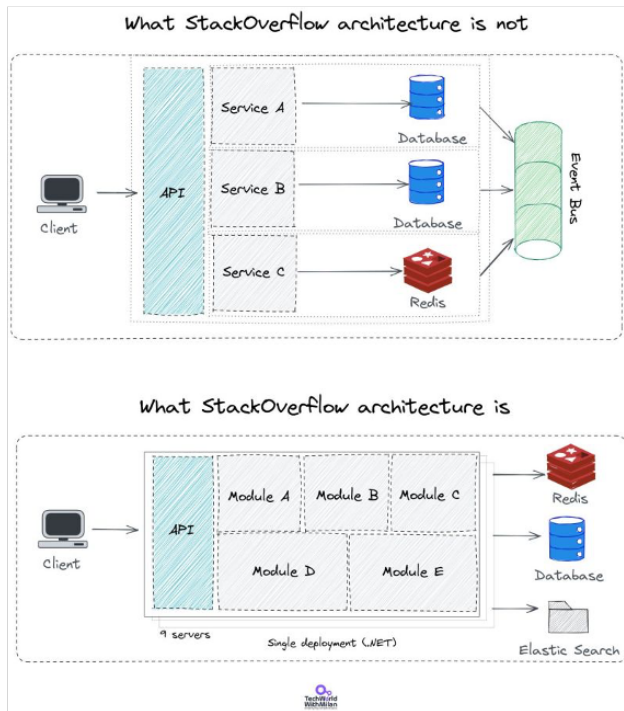
ByteByteGo
By Alex Xu



Learning resources for future Software Architects



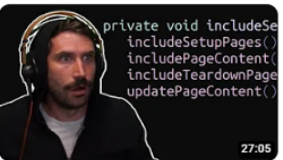
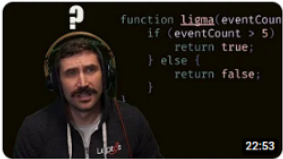
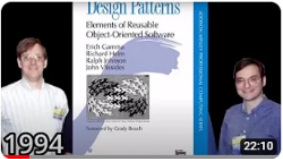
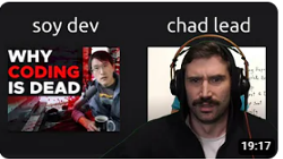
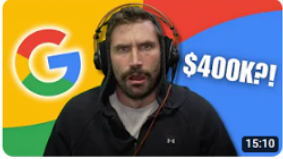
TechWorldWithMilan
Dr Milan Milanović



Learning resources for future Software Architects



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Learning resources for future Software Architects



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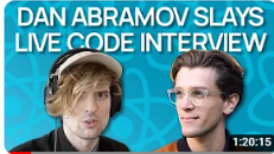
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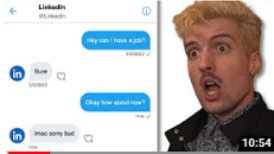
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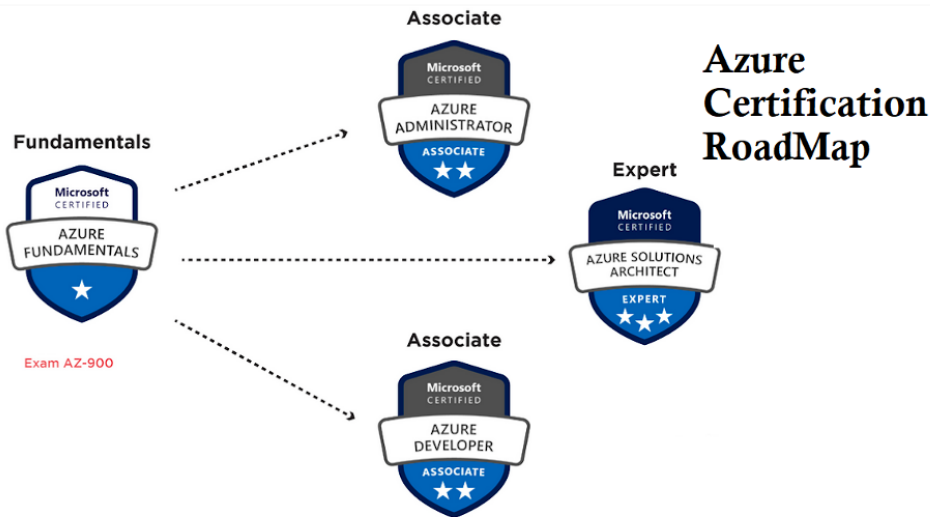


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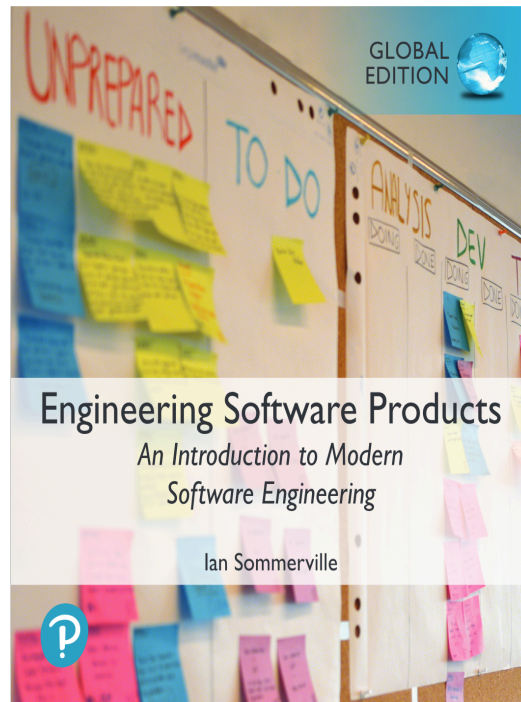
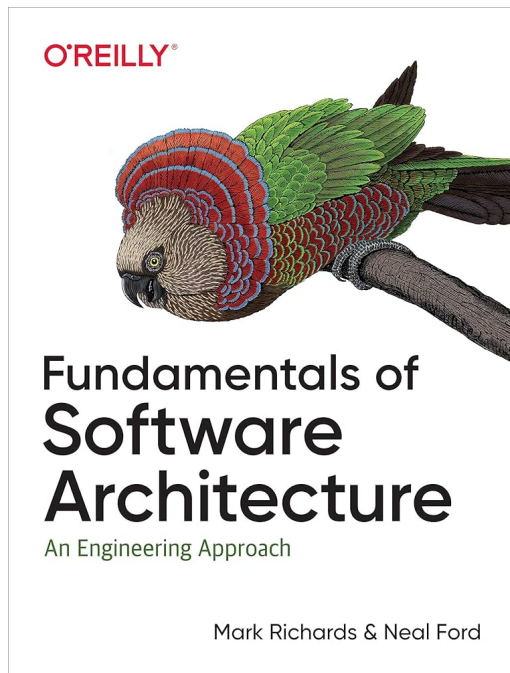
Courses and Certifications for Software Architects



Nu vă recomand să cheltuiți banii voștri pentru astfel de certificări.
Mulți angajatori oferă în oferta de angajare accesul la cursuri.

Books

Books aren't for learning tools/languages/syntax.
Books are for understanding patterns of thought.



Kahoot!

Next time on ...

Course 7

Please read from
Fundamentals of Software Architecture (Mark Richards & Neal Ford):

- Chapter 19 – Architecture Decisions (separated PDF in Teams)

Engineering Software Products (Ian Sommerville)

- Chapter 4 - 4.5 Technology Issues

